A GEODESIC-PRESERVING MAP OF A EUCLIDEAN CONE DISK IS AFFINE

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Abstract. In this paper, we solve a rigidity problem on geodesic-preserving maps of the unit disk $\mathbb{D}$ into itself. The unit disk $\mathbb{D}$ is equipped with a Euclidean cone metric which is induced by the lift of a holomorphic quadratic differential on a compact Riemann surface. The main result is that an injective map of the disk into itself is an affine homeomorphism if it is geodesic-preserving.

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1. Introduction

Let $\mathbb{R}^n$ be the n-dimensional Euclidean space. A map $f : \mathbb{R}^n \mapsto \mathbb{R}^n$ is called geodesic-preserving if the image of a geodesic under f is a geodesic. It is a classical result in affine geometry that a geodesic-preserving map of the n-dimensional Euclidean space is affine. Many mathematicians studied this kind of maps comprehensively. For example, in conformal geometry, Beardon and Minda proved the following result.

Theorem 1 ([1])**.** *A map $f : \hat{\mathbb{R}}^n \mapsto \hat{\mathbb{R}}^n$ is a Möbius transformation if and only if it is a locally sphere-preserving map.*

In hyperbolic geometry, the following result is due to Li and Wang.

Theorem 2 ([7])**.** *Suppose that $f : \mathbb{D}^n \mapsto \mathbb{D}^n$ is injective and maps any r-hyperplane into some r-hyperplane for some positive integer $r < n$. Then it is an isometry or a composition of an isometry and an affine transformation under the Klein model if and only if f is non-degenerate.*

There are still many other generalizations. For a more detailed list of such kind of rigidity results, we refer the reader to [7]. In this paper, we study this problem in the unit disk with a Euclidean cone metric induced by the lift of a holomorphic quadratic differential on a compact Riemann surface. The following is our main theorem:

Theorem 3. *Let ϕ be a holomorphic quadratic differential on a compact Riemann surface of genus $g \geq 2$ and $\tilde{\phi}$ be its lift to the unit disk $\mathbb{D}$. Let $\mathbb{D}$ be equipped with the flat metric $|\tilde{\phi}|$. Suppose that $f : \mathbb{D} \mapsto \mathbb{D}$ is injective and geodesic-preserving. Then f is affine.*

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2. Background

In this section, we recall some basic knowledge about holomorphic quadratic differentials and the related geometry.

2.1. **The local geometry of quadratic differentials.** Given any holomorphic quadratic differential ϕ on a Riemann surface X, at every regular point $p \in X$, there exists a local coordinate z on a sufficiently small neighborhood U of p such that $\phi = dz^2$ on U. The coordinate z is given by the integral $z(y) = \int_p^y \sqrt{\phi}$. At every singular point q, there exists a local coordinate z in a small neighborhood V such that $\phi = z^k dz^2$ in V, where k is the order of ϕ at q. These are called the natural coordinates of holomorphic quadratic differentials. For a detailed introduction, we refer the reader to [10]. Every holomorphic quadratic differential induces a metric that is singular at zeroes. The local behavior of this metric in a neighborhood of a regular point is exactly Euclidean. In a small neighborhood of a singular point of order k, this metric is obtained by patching together $k+2$ Euclidean half-spaces, as shown in the following picture.

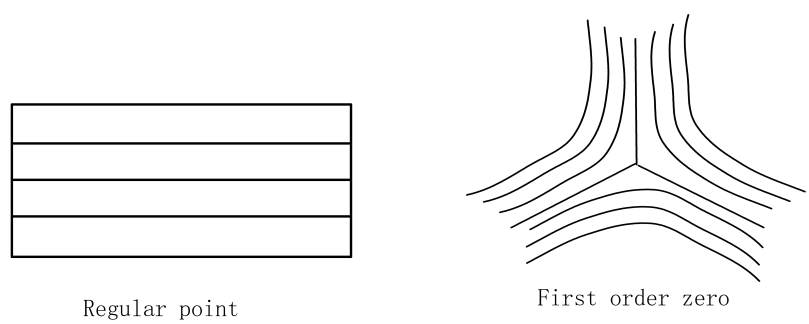

Figure 1. The local geometry of a holomorphic quadratic differential

A θ-trajectory of ϕ is defined to be the maximal extension of

$$\{z \in \mathbb{D} : arg\phi(z) = \theta\}.$$

A geodesic segment is an isometric embedding of an interval $[a, b]$ into $\mathbb{D}$. It is made up of a sequence of trajectories. It's a classical result that two adjacent geodesic segments make up a new one if and only if the two angles they make are both greater than or equal to π. A geodesic is said to be non-singular if it doesn't contain any singular point. We always parameterize a geodesic (geodesic ray or geodesic segment) by length parameter and the choice of the base point should be clear from the context.

We make some conventions for terms and notations. We reserve the term geodesic to mean that it is of infinite length in both directions, and when we speak of a geodesic ray, we mean it is finite in one direction and infinite in the other. The two endpoints of a geodesic γ are denoted by $\gamma(+\infty)$ and $\gamma(-\infty)$ respectively. The endpoint on the boundary of a geodesic ray α is denoted by $\alpha(+\infty)$. The endpoints of a geodesic or geodesic ray are well defined by Theorem 5 below.

Remark. We follow the convention that at any singular point p of order k, the cone angle at p is defined to be $(k+2)\pi$. Other authors may use the intrinsic conformal angle which differs from ours by a factor $\frac{2}{k+2}$.

2.2. **The large-scale geometry of quadratic differentials.**

Teichmüller's lemma dominates the large-scale behavior of quadratic differentials.

Theorem 4 (Teichmüller's lemma)**.** *Let G be the interior of a simple closed ϕ-polygon Λ with sides λ_i and interior angles θ_j at its vertices. The quadratic differential ϕ is supposed to be holomorphic in $\overline{G}$ (the only points on Λ which can be critical are the vertices). The zeroes of ϕ in G are denoted by z_i with n_i their orders. Then we have*

$$\sum_j (\pi - \theta_j) = 2\pi + \sum_i n_i \pi,$$

where n_i ranges over all zeroes of ϕ in G.

The following results are crucial for our proof.

Theorem 5 ([8])**.** *Let $\mathbb{D}$ be equipped with the metric $|\tilde{\phi}|$. Then the following results hold:*

1. *Any geodesic ray has a well-defined endpoint on $\partial\mathbb{D}$.*
2. *If γ_1, γ_2 are two geodesic rays emanating from a common origin O, then their endpoints are different.*
3. *If two geodesics γ_1, γ_2 share the same pair of endpoints on $\partial\mathbb{D}$, then they bound a Euclidean strip and the projection of γ_1, γ_2 are parallel simple loops.*

The large-scale geometry of $|\tilde{\phi}|$ behaves more like hyperbolic space despite that its local geometry is just Euclidean.

Definition 2.1. Two metric spaces (X_1, d_1) and (X_2, d_2) are called quasi-isometric if there exists a map f between X_1 and X_2 satisfying: there exist constants $\lambda \geq 1, \epsilon > 0$ and $K \geq 0$ such that for any $x, y \in X_1$,

$$\frac{1}{\lambda} d_1(x,y) - \epsilon \leq d_2(f(x), f(y)) \leq \lambda d_1(x,y) + \epsilon,$$

and every point of X_2 lies within a distance of K of the image $f(X_1)$.

Actually, since the fundamental group of a compact Riemann surface acts properly discontinuously and cocompactly on the unit disk $\mathbb{D}$, we can apply the classical Švarc-Milnor lemma:

Corollary 2.1. *The unit disk $\mathbb{D}$, when equipped with the metric induced by the lift of a holomorphic quadratic differential on a Riemann compact surface, is quasi-isometric to the hyperbolic space $\mathbb{D}_H$.*

The following is an immediate corollary of the above result which plays an important role in our proof:

Corollary 2.2. *There exists a constant K depending only on ϕ such that every geodesic γ in the $|\tilde{\phi}|$ metric is in a K-neighborhood of the hyperbolic geodesic connecting $\gamma(+\infty)$ and $\gamma(-\infty)$.*

In the proof of surjectivity of f, we make use of a homeomorphism between the boundary $\partial\mathbb{D}$ and the space of geodesic rays emanating from a fixed point. Fix any point $O \in \mathbb{D}$, if we equip the set of all geodesic rays $\mathcal{G}(O)$ emanating from O with the compact-open topology, then the boundary of the unit disk $\partial\mathbb{D}$ can be naturally identified $\mathcal{G}(O)$. Define a map $g : \mathcal{G}(O) \mapsto \partial\mathbb{D}$ in the following way: for any $\gamma \in \mathcal{G}(\mathcal{O})$, let $g(\gamma) = \gamma(+\infty)$. Then

Theorem 6. *The map g is a homeomorphism.*

For a detailed introduction of the above results, one can see [3, Chapter D.3] and [9, Chapter 10]

3. Proof of the main result

From now on, we always equip $\mathbb{D}$ with the metric $|\tilde{\phi}|$ unless otherwise stated. When we speak of the map f, we mean the one defined in Theorem 3.

Definition 3.1. A map of $\mathbb{D}$ into itself is called geodesic-preserving if the image of a geodesic is a geodesic.

Definition 3.2. A geodesic chain is a finite sequence of geodesics $\{\gamma_k\}_{1\leq k\leq n}$ such that $\gamma_k \cap \gamma_{k+1} \neq \emptyset$ for $1 \leq k < n$. Two disjoint geodesics α, β are said to be connected by a geodesic chain Γ if $\Gamma \cap \alpha \neq \emptyset$ and $\Gamma \cap \beta \neq \emptyset$.

3.1. The map f is continuous on geodesics.

Proposition 3.1. *Suppose that γ is a geodesic, and $p, q \in \gamma$. Suppose that α, β are two geodesic rays emanating from p, q respectively and lying on the same side of γ. The two angles formed by $\overline{pq}$ and α , $\overline{qp}$ and β are denoted by $\angle P$ and $\angle Q$ respectively, as shown in Figure 2. If $\angle P + \angle Q > \pi$, then the endpoints of α and β are different.*

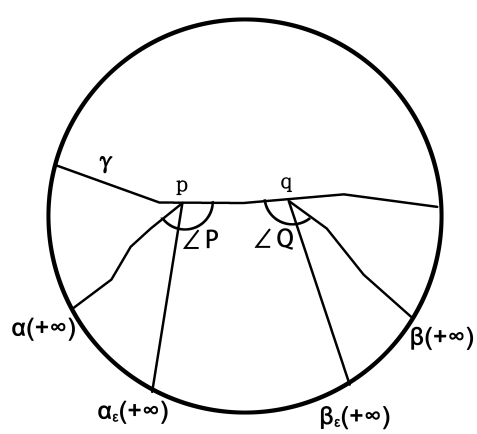

Figure 2. Two rays with different endpoints

Proof. By Teichmüller's lemma, the two rays α and β are disjoint. Without loss of generality, we can assume that α is on the left side of β. Take a geodesic ray α_ϵ emanating from p and on the right of α such that the angle formed by α_ϵ and α is smaller than ϵ. By Theorem 5, its endpoint is different from $\alpha(+\infty)$. In the same way, we get a geodesic ray β_ϵ on the left of β. Taking ϵ sufficiently small, we can make sure that α_ϵ and β_ϵ are disjoint. Therefore, the two endpoints $\alpha(+\infty)$ and $\beta(+\infty)$ are separated by $\alpha_\epsilon(+\infty)$ and $\beta_\epsilon(+\infty)$. Obviously, they are different. □

The following lemma is the core of our proof.

Lemma 3.1. *The restriction of f to a geodesic γ — $f|_\gamma$ is order-preserving.*

We first prove that if $A, B, C \in \gamma$ and B is in the middle of A and C, then $f(B)$ is also in the middle of $f(A)$ and $f(C)$. A map $g : \mathbb{R} \mapsto \mathbb{R}$ satisfying this condition is said to have the middle-point-preserving property.

Proposition 3.2. *There exist three geodesics γ_A, γ_B and γ_C satisfying the following conditions:*

(1) *These three geodesics are pairwise disjoint.*
(2) *$\gamma_A(0) = A, \gamma_B(0) = B$ and $\gamma_C(0) = C$. The geodesic γ_A has the property that the two subrays $\gamma_A(0, +\infty)$ and $\gamma_A(0, -\infty)$ lie on different sides of γ, and so do γ_B and γ_C.*
(3) *$\gamma_A(+\infty), \gamma_B(+\infty)$ and $\gamma_C(+\infty)$ lie on the same side of γ.*
(4) *$\gamma_A(+\infty), \gamma_B(+\infty), \gamma_C(+\infty)$ and the two endpoints of γ are pairwise different.*

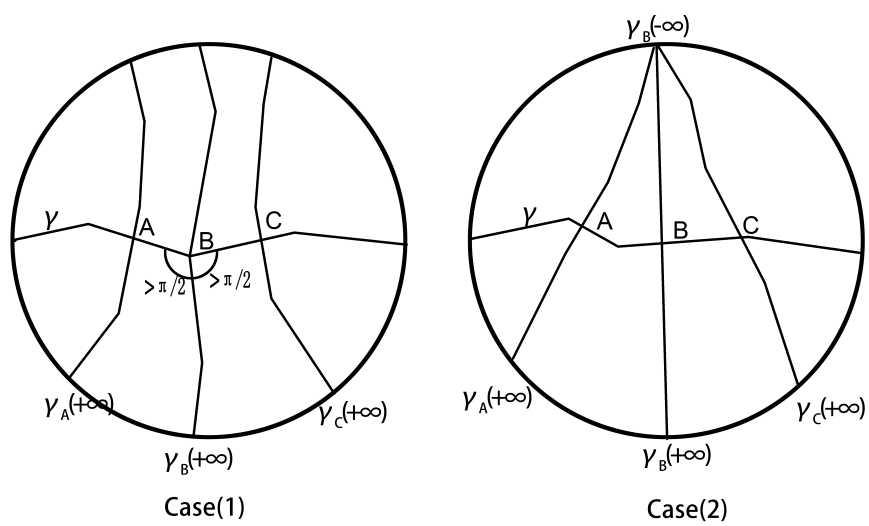

FIGURE 3. Three geodesics with different endpoints

Proof. Since γ is a geodesic, the two angles at B formed by γ are all greater than or equal to π.

If B is a singular point, extend the angular bisector of each angle formed by γ at B to the boundary. Obviously, the composition of these two angular bisectors is geodesic which we denote by γ_B. We can assume $\gamma_B(0) = B$. Suppose the angle at B formed by γ on its bottom side is strictly greater than π, and the ray $\gamma_B(0, +\infty)$ lies on this side. We construct γ_A at A and γ_C at C in the same way. Then by Proposition 3.1, the three geodesics γ_A, γ_B and γ_C are pairwise disjoint and their endpoints on the bottom side of γ are all different. See Figure 3 case(1).

If B is a normal point, consider the family of geodesics through B. Any two of them intersect at a single point. Since there are only countably many periodic directions and countably many singular points, almost every geodesic in this family is non-singular and non-periodic. Choose a non-singular and non-periodic geodesic from this family, and denote it by γ_B. Take the geodesic ray connecting $\gamma_B(-\infty)$ and A. Make it cross γ and tend to the boundary. We get a geodesic passing through A and denote it by γ_A. Similarly we get γ_C. By Theorem 5, these three geodesics are pairwise disjoint and their endpoints $\gamma_A(+\infty), \gamma(+\infty)$ and $\gamma_C(+\infty)$ are all different. See Figure 3 case(2). □

The two endpoints of γ divide the boundary into two parts. From now on, we'll work in the part which $\gamma_A(+\infty), \gamma_B(+\infty)$ and $\gamma_C(+\infty)$ lie on. In addition, we assume that $\gamma_A(+\infty), \gamma_B(+\infty)$ and $\gamma_C(+\infty)$ are in counter-clockwise order.

Proposition 3.3. *There are two geodesic chains* $\Gamma_\alpha = \{\alpha_1, ..\alpha_n\}$ *and* $\Gamma_\beta = \{\beta_1, ..\beta_m\}$ *satisfying the following conditions:*

(1) $\Gamma_\alpha \cap \gamma_A \neq \emptyset, \Gamma_\alpha \cap \gamma_B \neq \emptyset$ *and* $\Gamma_\alpha \cap \gamma_C = \emptyset$.
(2) $\Gamma_\beta \cap \gamma_B \neq \emptyset, \Gamma_\beta \cap \gamma_C \neq \emptyset$ *and* $\Gamma_\beta \cap \gamma_A = \emptyset$.
(3) $\Gamma_\alpha \cap \gamma = \emptyset, \Gamma_\beta \cap \gamma = \emptyset$.
(4) $\Gamma_\alpha \cap \Gamma_\beta \neq \emptyset$.

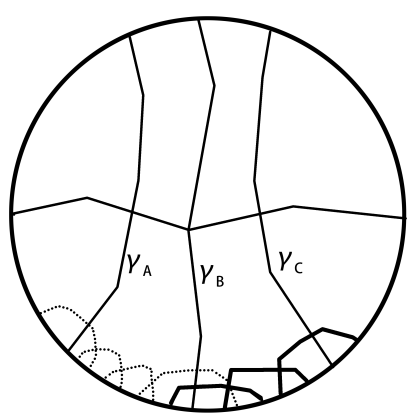

FIGURE 4. Geodesic chains

Proof. Since $\mathbb{D}$ is quasi-isometric to the hyperbolic plane $\mathbb{D}_H$, every point $x \in (\gamma_A(+\infty), \gamma_B(+\infty))$ has a neighborhood U_x such that for any two points $p, q \in$

U_x, the geodesic connecting p, q is disjoint from γ and γ_C. By compactness, we get a finite sequence of intervals $\{(a_n, b_n)\}_{0\leq n\leq N}$ such that their endpoints $a_0, a_1, b_0, a_2, b_1 \dots a_n, a_{n+1}, b_{n-1}, a_{n+1}, b_n \dots$ lie on $\partial\mathbb{D}$ in counter-clockwise order, and $\gamma_A(+\infty) \in (a_0, b_0)$, and $\gamma_B(+\infty) \in (a_N, b_N)$. For each n, take the geodesic connecting a_n and b_n. Then these geodesics make up a geodesic chain which we denote by Γ_α. Similarly, for the two endpoints $\gamma_B(+\infty)$ and $\gamma_C(+\infty)$, we get a geodesic chain Γ_β. One can easily check that Γ_α and Γ_β satisfy these conditions above. In Figure 4, the dotted geodesics form a geodesic chain connecting γ_A and γ_B, and the bold geodesics form a geodesic chain connecting γ_B and γ_C. □

Proposition 3.4. *Let $g : \mathbb{R} \mapsto \mathbb{R}$ be an injective map. If g has the middle-point-preserving property, then g is monotone.*

Proof. The proof is quite elementary. Up to a homeomorphism of $\mathbb{R}$, we assume that $g(0) < g(1)$. We now show that g is strictly increasing. Take two points $x, y \in \mathbb{R}$ and suppose that $x < y < 0$. Since 0 is in the middle of y and 1, $g(0)$ is in the middle of $g(y)$ and $g(1)$. Then it follows that $g(y) < g(0) < g(1)$. Since y is in the middle of x and 0, it follows that $g(x) < g(y) < g(0)$.

The other cases are similar. We omit the details. □

Any geodesic γ divides the unit disk $\mathbb{D}$ into two parts. Denote these two parts by $\mathbb{D}^+_\gamma$ and $\mathbb{D}^-_\gamma$.

Proof of Lemma 3.1. By Proposition 3.4, we have only to prove that f has the middle-point-preserving property. We argue by contradiction. Suppose that $f(A)$ is in the middle of $f(B)$ and $f(C)$.

If $f(\gamma_A)$ lies totally on the half-space $\mathbb{D}^+_{f(\gamma)}$, then the geodesic chain $f(\Gamma_\alpha)$ also lies totally on $\mathbb{D}^+_{f(\gamma)}$ since it intersects $f(\gamma_A)$ and doesn't intersect $f(\gamma)$. Similarly, since $f(\Gamma_\beta)$ intersects $f(\Gamma_\alpha)$ and doesn't intersect $f(\gamma)$, $f(\Gamma_\beta)$ lies totally on $\mathbb{D}^+_{f(\gamma)}$. Take $P \in \gamma_B \cap \Gamma_\beta$ and $Q \in \gamma_C \cap \Gamma_\beta$. Observe that $f(\gamma_A)$ divides $\mathbb{D}^+_{f(\gamma)}$ into three parts. If two points $f(P)$ and $f(Q)$ lie on the same part, then either $f(\gamma_B)$ or $f(\gamma_C)$ intersects $f(\gamma_A)$ or $f(\gamma)$. If not, $f(\Gamma_\beta)$ intersects $f(\gamma_A)$ or $f(\gamma)$. Anyway, we get a contradiction.

Otherwise, the two geodesics $f(\gamma_B)$ and $f(\gamma_C)$ must lie on different sides of $f(\gamma_A)$. Consider the geodesic chain $f(\Gamma_\beta)$. It intersects both $f(\gamma_B)$ and $f(\gamma_C)$, and then it must intersect $f(\gamma_A)$. Again, we get a contradiction. □

Corollary 3.1. *For any geodesic γ, the restriction of f to $\gamma : f|_\gamma$ is a homeomorphism onto its image.*

Since any segment or ray can be extended to a geodesic, we get the following corollary:

Corollary 3.2. *The map f maps every geodesic segment homeomorphically onto a geodesic segment and every ray homeomorphically onto a ray.*

3.2. **The map f is surjective.**

Fix any point $O \in D$. The boundary $\partial\mathbb{D}$ can be naturally identified with all geodesic rays emanating from O. Since f maps every geodesic ray to a geodesic ray, it induces a map $\tilde{f}$ between $\partial\mathbb{D}$. For any point $x \in \partial\mathbb{D}$, define $\tilde{f}(x)$ to be the endpoint of $f(\gamma_x)$, where γ_x is the ray emanating from O and tending to x.

Since any two geodesic rays emanating from the same point never share the same endpoint, $\tilde{f}$ is injective.

Lemma 3.2. *The map $\tilde{f}$ is continuous.*

Proof. Fix a point $x_0 \in \partial\mathbb{D}$ and a small neighborhood $(a, b) \subset \partial\mathbb{D}$ of x_0. Take a sequence of intervals (a_n, b_n) satisfying the following conditions:

(1) $x_0 = \cap_{n=0}^{+\infty}(a_n, b_n)$.
(2) $(a, b) = (a_0, b_0) \supset (a_1, b_1) \supset ... \supset (a_n, b_n) \supset (a_{n+1}, b_{n+1}) \supset ...$
(3) $a_n \neq a_{n+1}$ and $b_n \neq b_{n+1}$ for each n.
(4) The geodesic γ_n connecting a_n and b_n is non-singular for each n.

Only condition (4) needs some explanation. This is possible because of the following observation. Denote the ray connecting the two points $\{O, x_0\}$ by α. For each point $s \in \alpha$, take a geodesic γ_s orthogonal to α at s. Then we get a family of geodesics $\{\gamma_s\}_{s\in\alpha}$. By Teichmüller's lemma, these geodesics are pairwise disjoint. Since there are only countably many singular points, almost every geodesic in this family is non-singular.

We now show that $\tilde{f}(a_n)$ converges to $\tilde{f}(x_0)$. Denote the geodesic connecting a and b by γ. Since f is a homeomorphism on γ, $\tilde{f}$ is order-preserving on (a, b). Denote the ray connecting $\{O, a_n\}$ by α_n, and denote the geodesic connecting $\{a_k, b_k\}$ by γ_k. Then for any $k < n$, γ_k intersects α_n. Since γ_k is non-singular, they intersect at a single point p_{nk}. For any fixed k, p_{nk} converges to p_k where $p_k = \alpha \cap \gamma_k$. Since f is continuous on γ_k, the sequence $f(p_{nk})$ converges to f(p_k), which implies that $f(\alpha_n)$ converges to $f(\alpha)$ locally uniformly. Therefore, $\tilde{f}(a_n)$ converges to $\tilde{f}(x_0)$. For the same reason, $\tilde{f}(b_n)$ converges to $\tilde{f}(x_0)$. In addition with the property that $\tilde{f}$ is order-preserving on (a, b), we conclude f is continuous at x_0. □

Corollary 3.3. *The map $\tilde{f}$ is a homeomorphism.*

Proof. It is continuous, injective and proper. □

With the help of $\tilde{f}$, we can prove that f is surjective.

Corollary 3.4. *The map f is surjective.*

Proof. For any point $q \in \mathbb{D}$, take a geodesic ray α emanating from $f(O)$ and passing through q. Denote the preimage $\tilde{f}^{-1}(\alpha(+\infty))$ by x. The geodesic ray connecting $\{O, x\}$ is denoted by γ_x. By the definition of $\tilde{f}$, $f(\gamma_x)$ is the geodesic ray connecting $\tilde{f}(x)$ and $f(O)$, which is exactly α. Thus, γ_x is mapped homeomorphically onto α and q must have a preimage. □

3.3. **The map f is a homeomorphism.**

Lemma 3.3. *For any geodesic γ, if $p, q \in \mathbb{D}$ lie on the same side of γ, then $f(p)$ and $f(q)$ lie on the same side of $f(\gamma)$.*

Proof. Suppose that p, q lie on the same side of γ. Take two geodesic rays γ_p, γ_q emanating from p, q respectively. Assume that they are both disjoint from γ and their endpoints are away from the endpoints of γ. This is possible by the following observation. Fix a point $O \in \gamma$. Take two geodesic rays α and β emanating from O and passing through p and q respectively. Then the two subrays of α and β starting from p and q respectively satisfy our requirements. Take a geodesic chain

Γ_{pq} which is disjoint from γ and connects the two rays γ_p and γ_q. Then the image of Γ_{pq}, γ_p and γ_q lie on the same side of $f(\gamma)$, and thus the lemma follows. □

The above lemma implies that f maps a half-space into a half-space. In addition with surjectivity of f, we get the following corollary:

Corollary 3.5. *Given any geodesic γ, if there exists a point $p \in \mathbb{D}^+_\gamma$ such that $f(p) \in \mathbb{D}^+_{f(\gamma)}$, then f maps $\mathbb{D}^+_\gamma$ bijectively onto $\mathbb{D}^+_{f(\gamma)}$ and maps $\mathbb{D}^-_\gamma$ bijectively onto $\mathbb{D}^-_{f(\gamma)}$.*

Obviously, the same conclusion also holds for the closed half-spaces $\mathbb{D}^+_\gamma \cup \gamma$ and $\mathbb{D}^-_\gamma \cup \gamma$

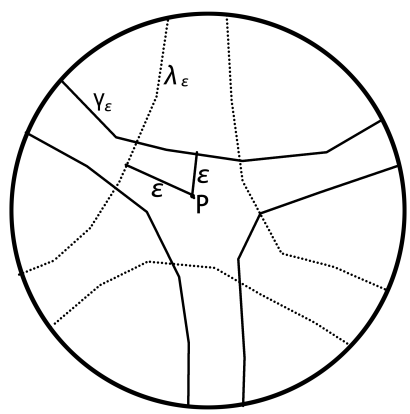

FIGURE 5. A small neighborhood bounded by horizontal and vertical geodesics

Lemma 3.4. *The map f is continuous.*

Proof. For any point $p \in \mathbb{D}$, construct a sequence of neighborhoods U_ϵ of p in the following way. Take all vertical trajectories at the distance ϵ from p, and denote them by $\{\gamma^\epsilon_n\}_{1\le n\le N}$ where $N-2$ is the order of p. Take all horizontal trajectories at the distance ϵ from p and denote them by $\{\lambda^\epsilon_n\}_{1\le n\le N}$. Then these horizontal and vertical trajectories bound a neighborhood U_ϵ of p. It is easily seen $U_\epsilon = \{\cap \mathbb{D}^+_{\gamma^\epsilon_n}\} \cap \{\cap \mathbb{D}^+_{\lambda^\epsilon_n}\}$ where $\mathbb{D}^+_{\gamma^\epsilon_n}$ or $\mathbb{D}^+_{\lambda^\epsilon_n}$ is the half-space containing p. See Figure 5. Take $\epsilon \to 0$, we get that $\bigcap_{\epsilon\to 0} f(\overline{U_\epsilon}) = f(\bigcap_{\epsilon\to 0} \overline{U_\epsilon}) = f(p)$. Then f is continuous at p. Since p is arbitrary, f is a continuous bijection. □

Corollary 3.6. *The map f is a homeomorphism.*

3.4. **The map f is affine.**

Theorem 3 now follows from a variation of the classical Pappus's Theorem. There's a detailed introduction to this theorem and its generalizations in [7]. For completeness, we include a brief proof here of what we need.

Lemma 3.5. *Given a triangle $\triangle OP_3Q_3$ on the Euclidean plane, suppose that there are two pairs of points $\{P_1, P_2\}$, and that $\{Q_1, Q_2\}$ on OP_3 and OQ_3 respectively, as shown in Figure 6. Fix the two lines P_1Q_1 and P_2Q_2. Move Q_3 along the segment Q_3P_3 towards P_3. During this process, suppose that P_1Q_2, P_2Q_1 intersect at M and P_2Q_3, P_3Q_2 intersect at N. Then P_1Q_1, P_2Q_2 and P_3Q_3 are parallel if and only if the three points $\{O, M, N\}$ are always colinear when we move Q_3 towards P_3.*

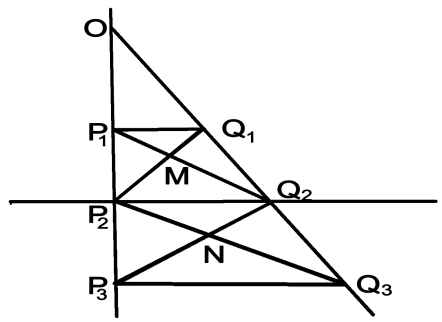

FIGURE 6. Pappus's Theorem

Proof. Up to an affine map of $\mathbb{R}^2$, we can assume that $P_2 = (0,0), O = (0,1), P_1 = (0,a), P_3 = (0,b)$ and $Q_2 = (c,0)$. Denote the slopes of P_1Q_1 and P_3Q_3 by k_1 and k_2 respectively. The coordinate of a point Z is represented by (x_Z, y_Z). By a direct computation, we get the coordinates of M and N.

$$\begin{cases} (x_M, y_M) = (\dfrac{(1-a)ac}{k_1c + 2a - a^2}, \dfrac{k_1ac + a^2}{k_1c + 2a - a^2}) \\ (x_N, y_N) = (\dfrac{(1-b)bc}{k_2c + 2b - b^2}, \dfrac{k_2bc + b^2}{k_2c + 2b - b^2}) \end{cases}$$

An elementary computation shows that $OM /\!/ ON$ for any c if and only if $k_1 = k_2 = 0$. The other direction is obvious. □

Proof of Theorem 3. The above lemma implies that f maps parallel geodesics to parallel geodesics. Fix a square small $Q \subset \mathbb{D}$. The central point of Q is denoted by O and the horizontal and vertical geodesics through O are denoted by α and β respectively. Up to an affine map, we may assume that f maps Q homeomorphically onto itself and fixes the four vertexes of Q. Then it necessarily fixes the central point O, and thus leaves α and β invariant. Observe that α and β divide Q into four squares. We can see that each $\frac{1}{4}$ subsquare of Q is invariant under f. By an inductive argument, we see that f is the identity map on Q. We complete the proof of Theorem 3. □

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